Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Paik, Ho Jung Ph.D.		
Project Title:	Short-Range Inverse-Square Law Experiment in Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hpaik@physics.umd.edu	**Fax:**	FY 301-405-6087
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-6086
Organization Name:	University of Maryland		
PI Address 1:	Department of Physics, University of Maryland		
PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Paik, Ho Jung		
COI Name (Institution):	Moody, M. (University of Maryland)		
Grant/Contract No.:	NAG3-2874		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth"s gravity.		
	The test Newton's law of gravitation with a resolution one million times better than the existing limits at 0.1 mm, the superconducting accelerometer technology will be combined with advantages of the low-g environment of space. The experiment will be cooled to < 2 K and the test masses will be magnetically suspended. The soft, low-loss suspension, combined with a low-noise SQUID, leads to extremely low intrinsic noise in the detector. To minimize Newtonian errors, the experiment employs a near null source, a circular disk of large diameter-to-thickness ratio. Two test masses, also disk-shaped, are suspended on the two sides of the source mass at a distance of 0.1 mm and form a superconducting differential accelerometer. A ground test apparatus is under construction and a laboratory test of the inverse-square law is planned. The objective of ISLES (Inverse-Square Law Experiment in Space) is to perform a null test of Newton's law on ISS with a resolution of 10 ppm at distances ranging from 100 micrometer to 1 mm. This		

Task Description:	represents an improvement by 10^6 over the existing limits at 100 micrometers, and even greater improvement at shorter distances. ISLES will be sensitive enough to detect axions with the strongest allowed coupling and test the string-theory prediction with R > 5 micrometer. To reach this sensitivity, the existing superconducting accelerometer technology will be combined with advantages of the low-g environment of space. The Standard Model of particle physics, although highly successful, has one serious blemish: the so-called "strong CP problem." A very plausible explanation for this phenomenon was proposed over two decades ago, but experimentalists have so far failed to verify it. One ramification of the theory is the prediction of the existence of a new particle, called the "axion," which mediates a short-range mass-mass interaction, as well as a force between intrinsic spin and mass. The axion is also a strong candidate for cold dark matter. Confirmation (or rejection) of this prediction would therefore have a major impact on our understanding of the universe, from its most microscopic constituents to its grand structure. Another impetus for testing the law at the shortest ranges possible comes from string theories. In these theories, the extra spacetime dimensions are compactified, leaving the observed 3+1 spacetime dimensions. For two large extra dimensions, the law of gravity changes from 1/r^2 to 1/r^4, as the separation r is reduced to below the radius of compactification R. An intriguing possibility that R > 1 micrometer has been suggested recently that would imply a violation of the inverse-square law below 1 micrometer distance. This is the first prediction of a string theory that can be tested, and a discovery of this deviation would be ground breaking.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Much progress has been made in the analysis and construction of the ground apparatus. A detailed analysis has been carried out for source metrology errors to determine the required tolerances. The sensing circuit was fully analyzed to optimize its performance and define the inductances needed for the superconducting transformers. All the mechanical components have been machined, except for the source mass. Superconducting shields for the test masses have been fabricated by diffusion-bonding thin niobium sheets to niobium rings. Two 1.5-mm tantalum sheets have been "two face ground" to improve the surface flatness to better than 10 micrometers, to meet the stringent requirement. A capacitance-bridge thickness scanner for measuring the flatness has been constructed and is undergoing tests. The alignment coils for the instrument have been wound. We expect to complete the construction and assembly of the experimental hardware and start testing the apparatus over next year.
Bibliography Type:	Description: (Last Updated: 10/30/2007)
Abstracts for Journals and Proceedings	Paik, H. J.; Moody, M. V.; Prieto, V. A.; Strayer, D. M. "ISLES: probing extra dimensions and searching for the axion" in Proceedings of COSPAR 2004 Paris, France , Jul-2004
Articles in Peer-reviewed Journals	Paik, H. J.; Moody, M. V.; Strayer, D. M. "Short-range inverse-square law experiment in space" Gen. Rel. Grav. Vol. 36, 523-537 (2004) , Mar-2004
Papers from Meeting Proceedings	Paik, H. J.; Prieto, V.; Moody, M. V. "Probing extra dimensions using a superconducting accelerometer" in Proceedings of the 6th International Conference on Gravitation and Astrophysics, Seoul, Korea (October, 2003) Feb-2004
Presentation	Paik, H. J. "Probing extra dimensions using a superconducting gravity gradiometer" Seminar at California Institute of Technology Apr-2004
Presentation	Paik, H. J. "Does gravity experience extra dimensions?" Colloquium at Physics Department, California State University at Long Beach Sep-2004